Wavelet-Enhanced Deep Learning Ensemble for Accurate Stock Market Forecasting: A Case Study of Nifty 50 Index

Course : Machine Learning(CS308)

Subject Coordinator - Dr. Jignesh Bhatt

Presented by

Ankit Mishra (202311011) Alok Yadav (202311007) Priyanshu Maurya(202311067) Yanshu Patel(202311088)

Problem Statement

- Stock data is highly **unstable, noisy, and hard to predict**.
- **Single models** (LSTM, CNN, TCN) fail to capture all patterns.
- **Noise** in data lowers prediction accuracy.
- Past studies use **only one model** or **only denoising**, not both.
- No existing system combines **wavelet denoising + multiple deep models**.
- **Main Problem :** Build a **robust ensemble model** using **denoised data + deep learning models** for better prediction.

MOTIVATION

- Traditional models fail to capture complex patterns.
- Financial data is highly noisy and dynamic.
- Need for hybrid intelligent models.

Importance:

- Helps investors in decision-making.
- Reduces financial risks.
- Enables better forecasting.

LITERATURE REVIEW

- **Existing Approaches :**
  - ARIMA (Statistical model)
  - LSTM (Deep Learning)
  - CNN-based models
- **Limitations :**
  - Poor noise handling
  - Overfitting issues
  - Limited feature extraction

Proposed Approach :
Wavelet + Deep Learning Ensemble improves prediction accuracy.

Dataset Description

- **Features and Data Types:**
- **Start Date:** 2010-01-04
- **End Date:** 2025-12-29
- **Total Observations:** 3434
- **Total Attributes:** 4
- **Data Type:** Time-Series (Tabular)

Feature	Type
Open Price	Numerical
High Price	Numerical
Low Price	Numerical
Close Price	Numerical

- **Target Variable:** Close Price (Numerical)

- Real-world **highly volatile** stock index data
- Contains **noise**
- **Wavelet (Coif3) denoising** applied

Methodology

Step 1: Data Collection.

Step 2: Data Preprocessing.

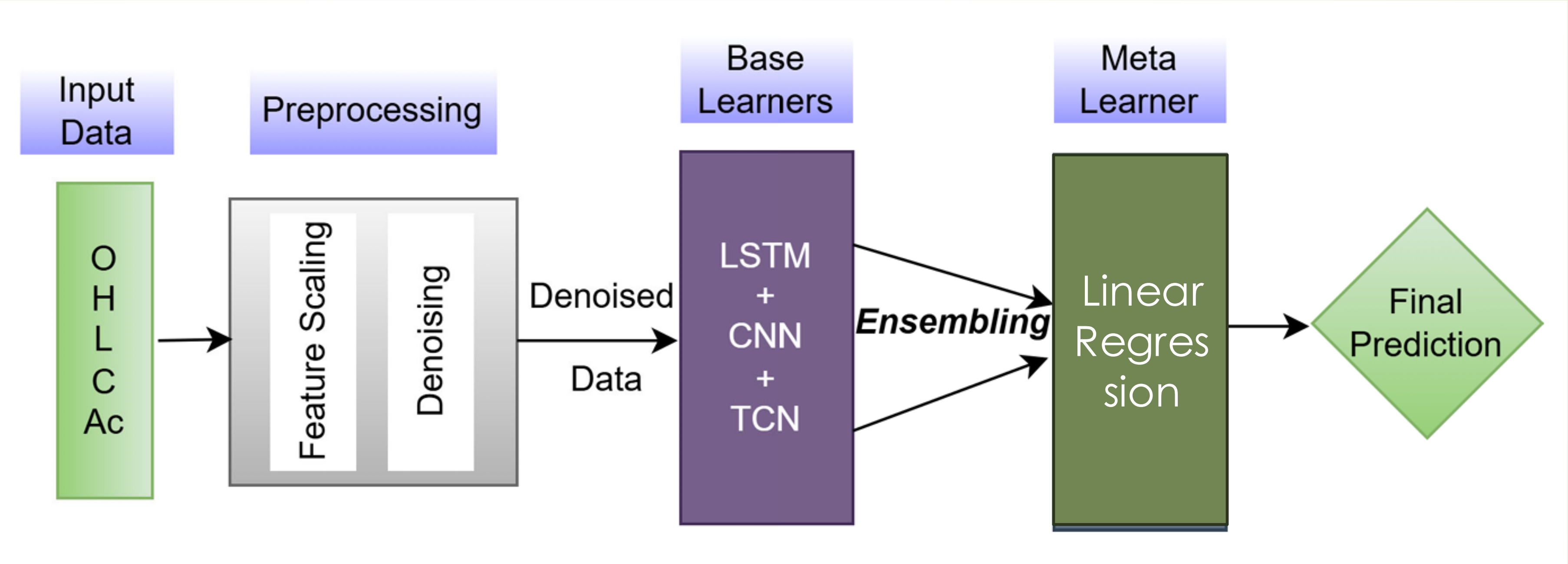
Step 3: Wavelet Transform (Noise Reduction).

Step 4: Model Training (LSTM).

Step 5: Ensemble Learning.

Step 6: Prediction & Evaluation.

System Architecture



WAVELET TRANSFORM

- Used for signal decomposition.
- Removes noise from stock data.
- Improves feature extraction.

Benefit:
Enhances model accuracy and stability.

MODEL USED

Model:

- LSTM (for time-series learning)
- CNN (for feature extraction)
- TCN (Temporal Convolution Network)
- Ensemble Model (Final prediction)

Why LSTM?

- Captures time-series dependencies.
- Handles sequential data effectively.

Why Ensemble?

- Combines strengths of all models,
- Gives best accuracy.

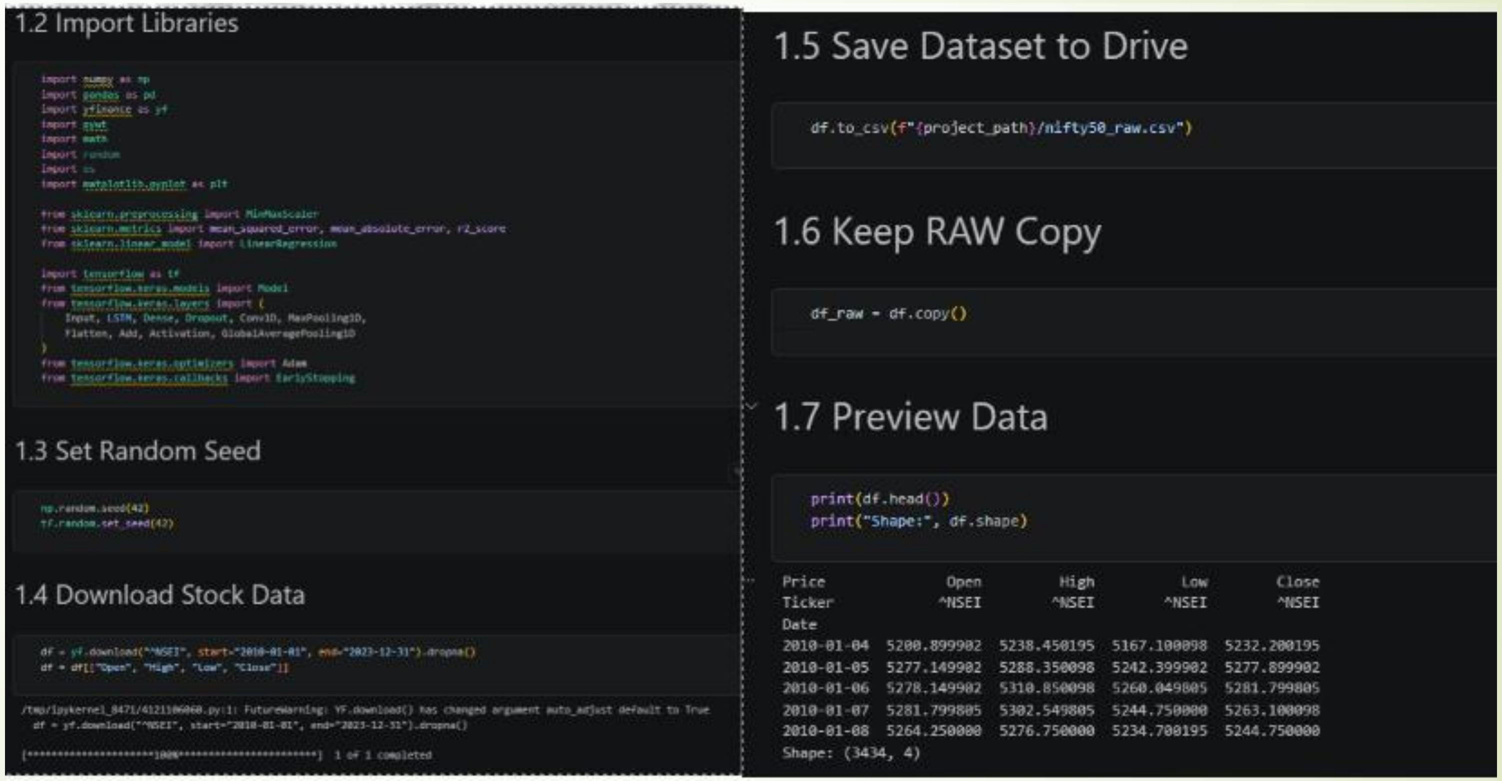
IMPLEMENTATION

Implementation Details:

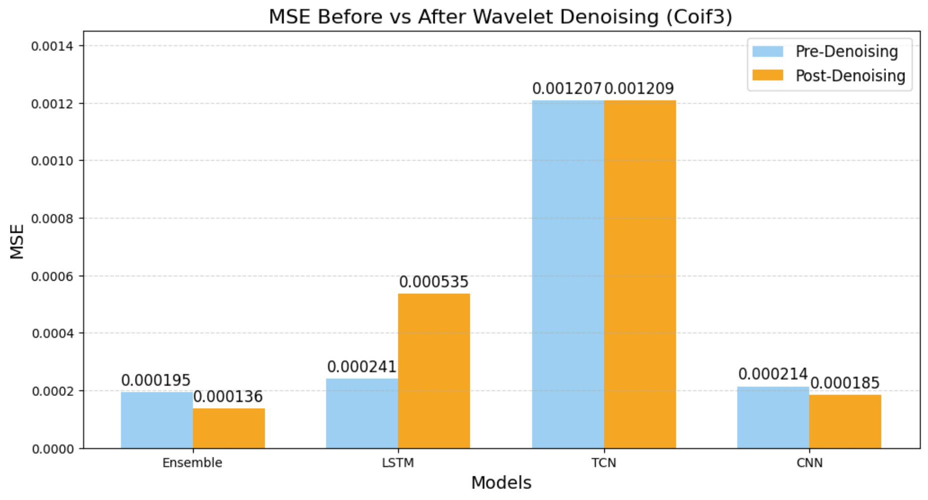
- Language: Python
- Libraries: NumPy, Pandas, TensorFlow, Keras, Matplotlib

Steps:

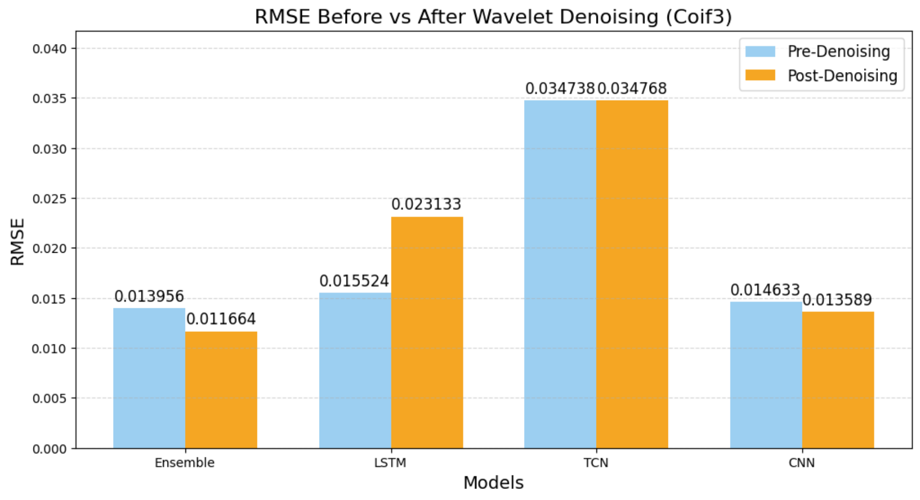
- Data Loading.
- Data Preprocessing.
- Model Training.
- Prediction.



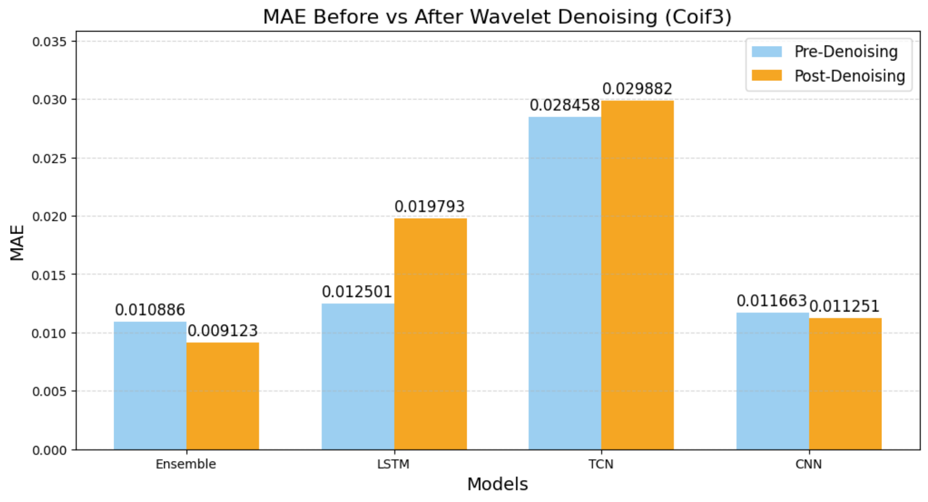
Results



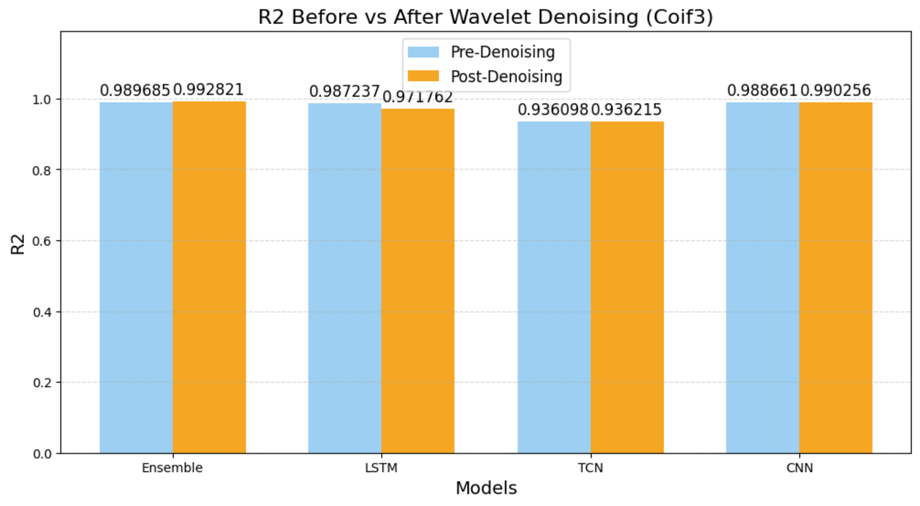
MSE Before vs After Wavelet Denoising (Coif3)
Pre-Denoising
Post-Denoising
MSE
Models
Ensemble
LSTM
TCN
CNN
0.000195
0.000136
0.000241
0.000535
0.001207
0.001209
0.000214
0.000185



RMSE Before vs After Wavelet Denoising (Coif3)
Pre-Denoising
Post-Denoising
RMSE
Models
Ensemble
LSTM
TCN
CNN
0.013956
0.011664
0.015524
0.023133
0.034738
0.034768
0.014633
0.013589



MAE Before vs After Wavelet Denoising (Coif3)
Pre-Denoising
Post-Denoising
MAE
Models
Ensemble
LSTM
TCN
CNN
0.010886
0.009123
0.012501
0.019793
0.028458
0.029882
0.011663
0.011251



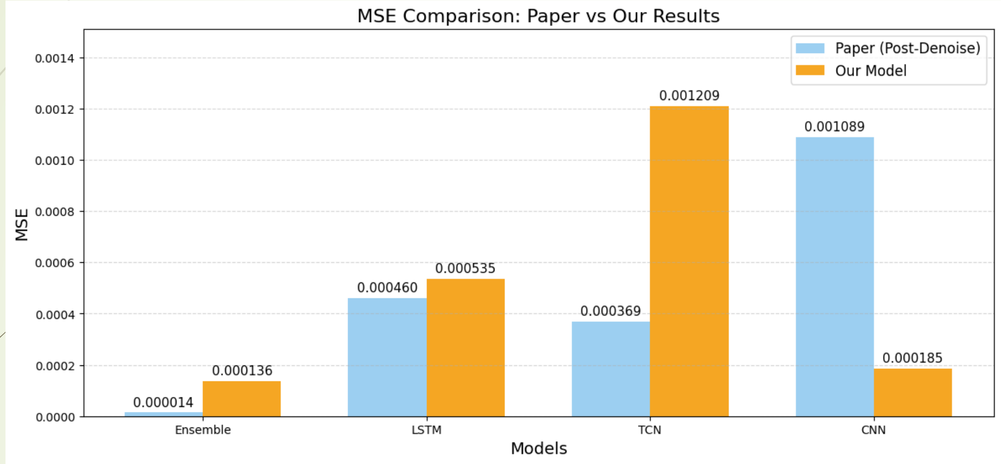
R2 Before vs After Wavelet Denoising (Coif3)
Pre-Denoising
Post-Denoising
R2
Models
Ensemble
LSTM
TCN
CNN
0.989685
0.992821
0.987237
0.971762
0.936098
0.936215
0.988661
0.990256

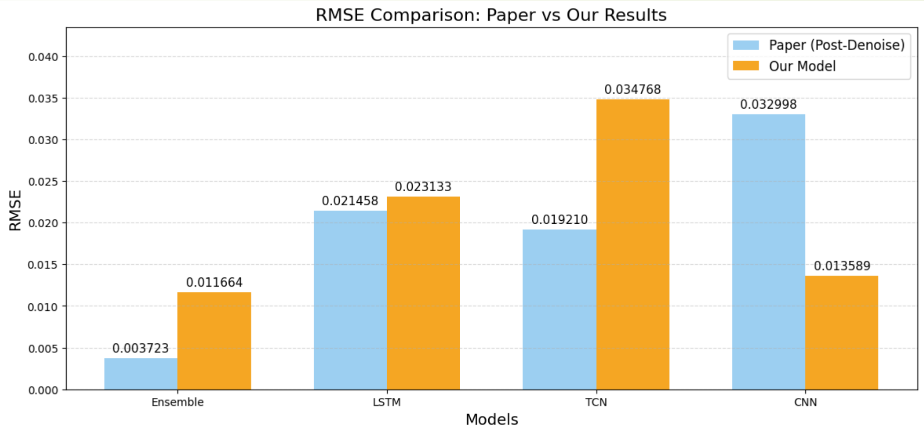
Delta changes

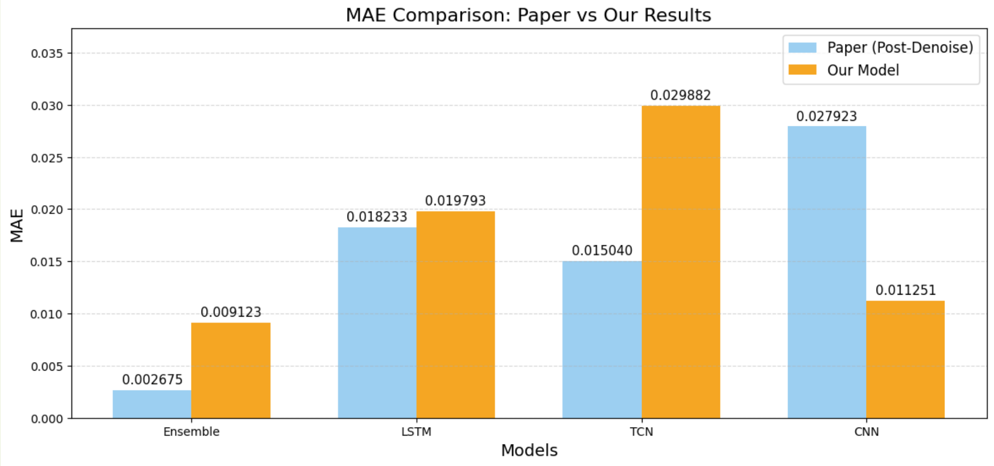
- Noise is present in all features, not just Close
- Cleaning all inputs gives a clearer signal and better model learning
- Our model predictions are **smooth**, so LR combines them better than other models
- LR is **more stable** and avoids overfitting on denoised data

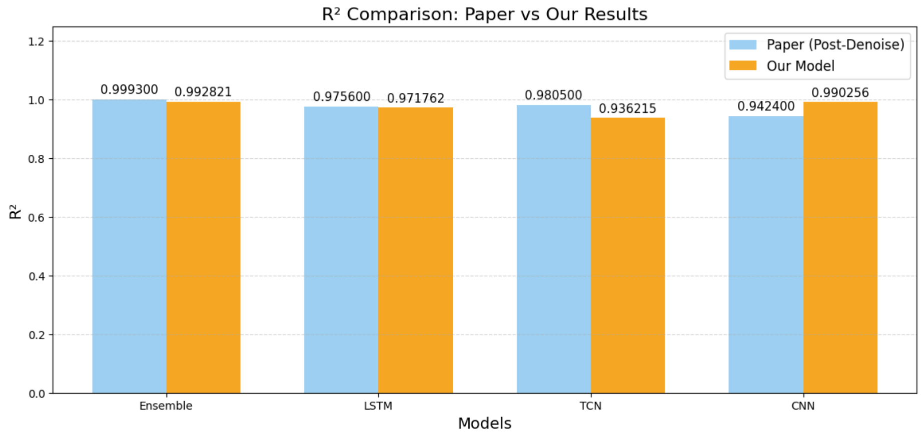
Category	Paper	Our Code
Denoising	Coif3 on Close only	Coif3 on all OHLC
Ensemble	Random Forest Stacking	Linear Regression

Delta changes Results









PERFORMANCE COMPARISON

Comparison with Traditional Models:

Traditional Models:

- Less accurate.
- Sensitive to noise.

Proposed Model:

Higher accuracy.

Better noise handling.

Stable predictions.

Best Model:

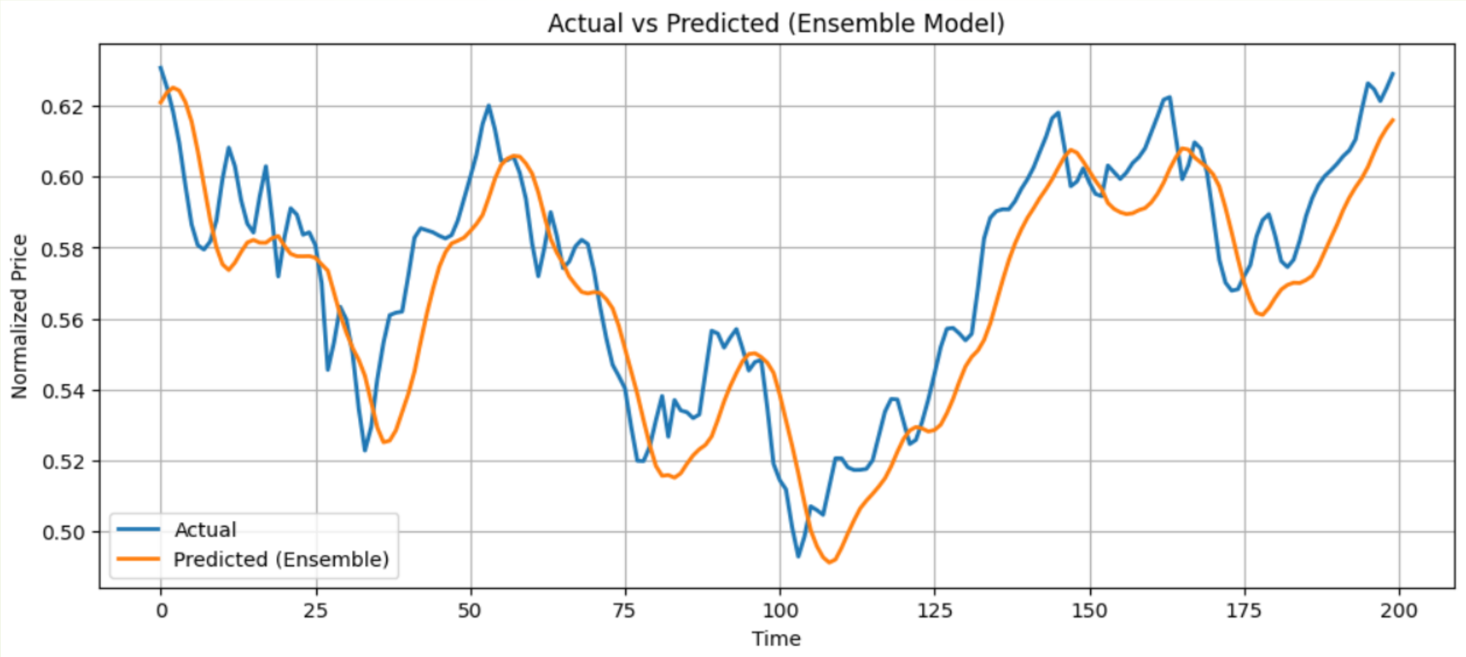
- LSTM gives best performance.
- Ensemble provides stable results.

CONCLUSION

Wavelet-Enhanced Deep Learning Ensemble significantly improves stock price prediction.

Applications:

- Financial forecasting.
- Investment decision-making.
- Risk management.



REFERENCES

- IEEE Research Paper:

  https://ieeexplore.ieee.org/document/11005722

Wavelet-Enhanced Deep Learning Ensemble for Accurate Stock Market Forecasting: A Case Study of Nifty 50 Index

- Python Libraries Documentation.
- Yahoo finance Nifty 50 dataset source.